

Project Analysis Summary

1. Data Exploration & Feature Selection

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Dataset includes **Sales** as the dependent variable and **TV, Radio, Social Media** as independent variables.

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Exploratory Data Analysis (EDA) confirmed distributions and relationships.

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All three predictors were initially included in the model.

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2. Multicollinearity Check

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Correlation matrix and **Variance Inflation Factor (VIF)** were used.

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No severe multicollinearity detected (VIF values acceptable).

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Ensured predictors could be modeled together.

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3. OLS Model Building

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Built regression model:

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Sales~ TV+Radio+SocialMedia

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Reported **Adjusted R²**, **F-statistic**, and **p-values**.

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Stepwise regression used to drop insignificant predictors (e.g., Social Media if $p > 0.05$).

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4. Assumption Validation

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Linearity: Residuals vs. fitted plot showed no major deviations.

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Normality: Q-Q plot and histogram confirmed approximate normal distribution of residuals.

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Homoscedasticity: Residuals evenly spread across fitted values.

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Independence: Durbin-Watson statistic acceptable.

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5. Coefficient Interpretation

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TV: Positive, significant effect — each \$1K spent increases Sales by a measurable amount.

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Radio: Strongest ROI per \$1K spent — most efficient driver of Sales.

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Social Media: Not statistically significant — limited direct impact when controlling for TV and Radio.

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6. Business Recommendation

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Prioritize Radio: Highest ROI, strongest statistical evidence.

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Maintain TV: Reliable contributor to Sales growth.

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Reduce Social Media spend: Reallocate budget toward Radio and TV until stronger evidence emerges.

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Strategic roadmap: Short-term reallocation, medium-term reassessment, long-term regression-based optimization.

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